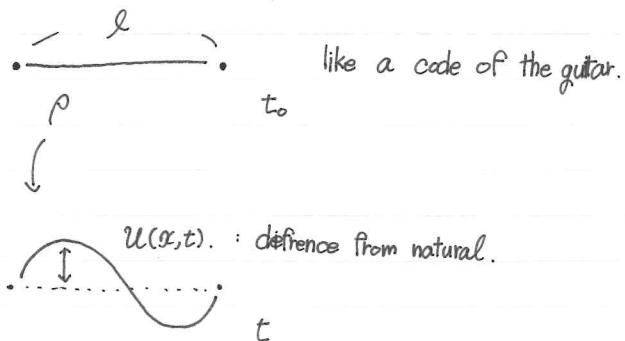


Chapter 6. wave eq.

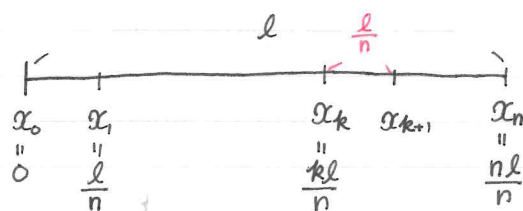
Chapter 6. wave eq.

density $\rho > 0$, length $l > 0$, constant $T > 0$.



<Kinetic E>

Kinetic $E := \frac{1}{2} \times (\text{mass}) \times (\text{velocity})^2$.



② (Velocity at x_k): $:= \frac{\partial}{\partial t} (x_k, t)$

③ (mass at x_k): $:= (\text{small length}) \times (\text{density})$

$$= \frac{l}{n} \rho.$$

$\therefore (\text{local kinetic } E) = \frac{1}{2} \times \frac{l}{n} \rho \times \left(\frac{\partial}{\partial t} (x_k, t) \right)^2$

summing up $k=0, \dots, n$

$$\sum_{k=0}^n \frac{1}{2} \frac{l}{n} \rho \left(\frac{\partial}{\partial t} (x_k, t) \right)^2 \sim \frac{1}{2} \int_0^l \rho \left(\frac{\partial}{\partial t} (x, t) \right)^2 dx$$

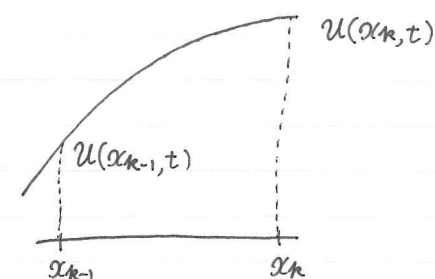
<Potential E>

Potential $E \rightarrow$ Fick's law

tension is ~~para~~ ^{比例} parallel to the variation rate (Δx)

of length of the string.
the

① (natural length) $= x_k - x_{k-1}$



② (variation of the length)

$$\doteq \sqrt{(x_k - x_{k-1})^2 + (u(x_k, t) - u(x_{k-1}, t))^2} - (x_k - x_{k-1})$$

$$= (x_k - x_{k-1}) \left(\sqrt{1 + \left(\frac{u(x_k, t) - u(x_{k-1}, t)}{x_k - x_{k-1}} \right)^2} - 1 \right)$$

↓ Taylor expansion $(1+\theta)^{\frac{1}{2}} \doteq 1 + \frac{1}{2}\theta + o(\theta^2)$.

$$\doteq (x_k - x_{k-1}) \left(\frac{1}{2} \cdot \left(\frac{u(x_k, t) - u(x_{k-1}, t)}{x_k - x_{k-1}} \right)^2 \right)$$

$n \rightarrow \infty$

$$\doteq (x_k - x_{k-1}) \left(\frac{1}{2} \frac{\partial^2 u}{\partial x^2} (x_k, t) \right)^2$$

(tension)

$$= \tau \times (\text{variation})$$

$$= \tau \times \sum_{k=1}^n (x_k - x_{k-1}) \cdot \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x}(x_k, t) \right)^2$$

$$\simeq \tau \times \int_0^l \frac{1}{2} \left(\frac{\partial u}{\partial x}(x, t) \right)^2 dx$$

Therefore

$$(\text{kinetic}) = \frac{\rho}{2} \int_0^l \left(\frac{\partial u}{\partial t}(x, t) \right)^2 dx$$

$$(\text{potential}) = \frac{\tau}{2} \int_0^l \left(\frac{\partial u}{\partial x}(x, t) \right)^2 dx$$

<Least action principle>

From Lagrange mechanics we define $\mathcal{L}$.

$$\mathcal{L} := (\text{kinetic } E) - (\text{potential } E)$$

Law of Equipartition of E エネルギー等分則.
phenomena turns to $(\text{kinetic } E) = (\text{potential } E)$ We define action S as

$$S = \int_0^T \mathcal{L} dt.$$

Least Action Principle 最小作用の原理.
 S が最小となる運動が実際に起こる.Define. *vec. func.*

$$S[u] := \int_0^T \left(\frac{\rho}{2} \int_0^l \left(\frac{\partial u}{\partial t}(x, t) \right)^2 dx - \frac{\tau}{2} \int_0^l \left(\frac{\partial u}{\partial x}(x, t) \right)^2 dx \right) dt.$$

 $\mathbb{Z}$ we want to know the minimizer of S .Let $\varphi := \varphi(x, t)$ $[0, l] \times [0, T]$ be a smooth func. For $\varepsilon > 0$ consider

$$S[u + \varepsilon \varphi] = f(\varepsilon)$$

If u is the minimizer of S then $S[u + \varepsilon \varphi] \geq S[u] = S[u + 0 \cdot \varphi]$.

$$\leadsto \frac{d}{d\varepsilon} \Big|_{\varepsilon=0} S[u + \varepsilon \varphi] = 0. \quad (6.1)$$

*everything
is smooth*Try to compute $\frac{d}{d\varepsilon} S[u + \varepsilon \varphi]$.

$$\begin{aligned} \frac{d}{d\varepsilon} S[u + \varepsilon \varphi] &= \frac{d}{d\varepsilon} \left(\int_0^T \left(\frac{\rho}{2} \int_0^l \left(\frac{\partial u}{\partial t}(x, t) + \varepsilon \frac{\partial \varphi}{\partial t}(x, t) \right)^2 dx - \frac{\tau}{2} \int_0^l \left(\frac{\partial u}{\partial x}(x, t) + \varepsilon \frac{\partial \varphi}{\partial x}(x, t) \right)^2 dx \right) dt \right) \\ &= \int_0^T \left(\frac{\rho}{2} \int_0^l \left(\frac{\partial u}{\partial t}(x, t) + \varepsilon \frac{\partial \varphi}{\partial t}(x, t) \right) \cdot \frac{\partial \varphi}{\partial t}(x, t) dx - \frac{\tau}{2} \int_0^l \left(\frac{\partial u}{\partial x}(x, t) + \varepsilon \frac{\partial \varphi}{\partial x}(x, t) \right) \frac{\partial \varphi}{\partial x}(x, t) dx \right) dt \end{aligned}$$

Letting $\varepsilon = 0$. together (6.1)

$$0 = \frac{d}{d\varepsilon} \Big|_{\varepsilon=0} S[u + \varepsilon \varphi].$$

$$= \int_0^T \left(\int_0^l \left(\rho \frac{\partial u}{\partial t}(x,t) \frac{\partial \varphi}{\partial t}(x,t) - \tau \frac{\partial u}{\partial x}(x,t) \frac{\partial \varphi}{\partial x}(x,t) \right) dx \right) dt$$

By integration by parts.

$$\int_0^T \int_0^l \rho \frac{\partial u}{\partial t}(x,t) \frac{\partial \varphi}{\partial t}(x,t) dx dt.$$

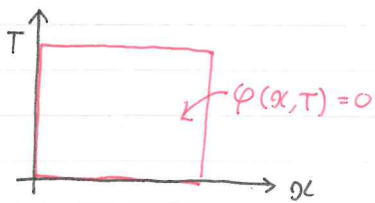
$$= \int_0^l \left(-\rho \int_0^T \frac{\partial^2 u}{\partial t^2}(x,t) \varphi(x,t) dt - \rho \left[\frac{\partial u}{\partial t}(x,t) \varphi(x,t) \right]_{t=0}^{t=T} \right) dx$$

$$\int_0^T \int_0^l \tau \frac{\partial u}{\partial x}(x,t) \frac{\partial \varphi}{\partial x}(x,t) dx dt$$

$$= \int_0^T \left(-\tau \int_0^l \frac{\partial^2 u}{\partial x^2}(x,t) \varphi(x,t) dx + \tau \left[\frac{\partial u}{\partial x}(x,t) \varphi(x,t) \right]_{x=0}^{x=l} \right) dt$$

Assume $\text{supp } \varphi \subset (0, l) \times (0, T)$ is cpt.

$$\left(\begin{array}{l} \varphi(0, T) = \varphi(l, T) = 0 \text{ For } t > 0 \\ \varphi(x, 0) = \varphi(x, T) = 0 \text{ For } 0 < x < l. \end{array} \right)$$



$$\int_0^l \left(\rho \frac{\partial^2 u}{\partial t^2} - \tau \frac{\partial^2 u}{\partial x^2} \right) \frac{\partial u}{\partial t} dx + \tau \left[\frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial t} \right]_{x=0}^{x=l} = 0 \quad \square$$

$O(E_g)$

Therefore

$$0 = \int_0^T \left(\int_0^l \left(-\rho \frac{\partial^2 u}{\partial t^2}(x,t) + \tau \frac{\partial^2 u}{\partial x^2}(x,t) \right) \varphi(x,t) dx \right) dt.$$

$\varphi \in C_0^\infty((0, l) \times (0, T))$ is arbitrary

$$\Rightarrow -\rho \frac{\partial^2 u}{\partial t^2}(x,t) + \tau \frac{\partial^2 u}{\partial x^2}(x,t) = 0.$$

$$\rho \frac{\partial^2 u}{\partial t^2}(x,t) = \tau \frac{\partial^2 u}{\partial x^2}(x,t) \quad (6.2)$$

is called (1-dim.) wave eq.

<Conservation law>

Prop 6.1: Let $\rho, \tau > 0$ be positive cont.

Let $u: [0, l] \times [0, T]$ be a sol. of

$$\begin{cases} \rho \frac{\partial^2 u}{\partial t^2}(x,t) = \tau \frac{\partial^2 u}{\partial x^2}(x,t) & 0 < x < l, t \in \mathbb{R} \\ \frac{\partial u}{\partial x}(x,t) = 0 & \text{at } x=0, l, t \in \mathbb{R} \end{cases}$$

Put

$$E[u] = \int_0^l \left(\underbrace{\frac{\rho}{2} \left(\frac{\partial u}{\partial t}(x,t) \right)^2}_{\text{kinetic}} + \underbrace{\frac{\tau}{2} \left(\frac{\partial u}{\partial x}(x,t) \right)^2}_{\text{position}} \right) dx$$

Then

$$\frac{d}{dt} E[u] = 0 \quad t \in \mathbb{R} \quad \leftarrow E[u] \text{ is independent of time.}$$

$$\text{Idea: } \int_0^l \frac{\partial u}{\partial t}(E_g) dx \Rightarrow \frac{d}{dt} E[u] = 0. \quad (E_g \rightarrow \text{Energy}).$$

$$\text{idea: Energy } E_g \quad \frac{d}{dt} E[u] = 0 \Rightarrow \int_0^l \frac{\partial u}{\partial t}(E_g) dx$$

$$\int_0^l \left(\rho \frac{\partial u}{\partial t} \cdot \frac{\partial^2 u}{\partial t^2} + \tau \frac{\partial u}{\partial t} \cdot \frac{\partial^2 u}{\partial x \partial t} \right) dx \quad \uparrow$$